

Theekshana Wijerathne

📍 Matale ✉ theekshana.22@cse.mrc.ac.lk ☎ 0761541254

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Summary

A highly motivated Computer Science and Engineering student with a strong focus on Software Engineering. Proficient in problem-solving, software design, and full-stack development, with a keen interest in building robust, scalable, and user-centered applications. Passionate about exploring modern development frameworks, tools, and best practices. Actively seeking a Software Engineering role to gain hands-on experience and contribute to the development of impactful, real-world software solutions.

Areas of Interest:

- Software Development (Front-end & Back-end)
- Database Systems
- Mobile Application Development
- DevOps & CI/CD Automation

Education

University of Moratuwa

March 2023 – present

Bachelor of Science (Hons) in Engineering, Computer Science and Engineering

- CGPA: 3.85 (Semester 1,2,3 Dean's list)

St.Thomas' college Matale

2013 – 2021

GCE Advanced Level- Physical Science Stream- 3A passes with a Z-score of 2.4718

Projects

Collaborative Learning Platform with personalized study plans

Jul 2025 – present

- building a microservices-based web platform for academic collaboration, enabling admins to manage workspaces for subjects, tutors to coordinate group learning, and students to engage through quizzes, collaborative notes, and personalized study plans. Features included resource tagging, real-time editing, whiteboards, and gamified progress tracking.
- **Contribution:** Full-stack development, System architecture, Deployment (Docker + CI/CD), Project coordination (Scrum Master)
- **Tech Stack:** React.js, Material UI, NestJS, PostgreSQL, TypeORM, Docker

Intelligent Advisor for Personal Finance & Investment



Apr 2025 – May 2025

Role: Full-Stack Developer & Scrum Master (Group 39)

- Developed an AI-powered financial advisory platform supporting stock prediction, budgeting, investment planning, and risk monitoring using integrated ML models and financial APIs.
- **Group 37:** Stock market prediction using LSTM and Yahoo Finance API.
- **Group 38:** AI-powered budgeting and expense tracking with NLP-based categorization (Scikit-learn).
- **Group 39 (My Contribution):** Built the personalized investment portfolio optimization module using the Markowitz Mean-Variance Model and Monte Carlo simulation. Developed REST APIs (FastAPI), frontend components (React.js), and managed PostgreSQL data operations. Led agile development and coordination as Scrum Master.
- **Group 40:** Financial risk and compliance monitoring using fairness-aware AI and explainable AI tools.
- **Tech Stack:** FastAPI, Next.js, PostgreSQL, YFinance API

Human Resource Management System

Aug 2024 – Oct 2024

- Developed a full-stack HRMS for Jupiter Apparels to streamline employee lifecycle management, leave request workflows, and HR operations. Key features included customizable job titles, role-based access control, and a responsive analytics dashboard. Focused on SQL query optimization, secure backend design, and fast, modular frontend development using Vite.
- **contribution** : Frontend, Backend, Database, Documentation
- **Tech Stack**: React.js (with Vite), Bootstrap, Node.js with Express, MySQL

Core Skills

- **Web Development**: HTML, CSS, React, Tailwind CSS, Node.js, Express, Next.js, FastAPI, NestJS, UI Libraries (Material UI, Bootstrap)
- **Mobile Development**: React Native
- **Database Management**: MySQL, MongoDB, PostgreSQL
- **Version Control**: Git, GitHub
- **Programming Languages**: C, C++, Python, Java, JavaScript, TypeScript
- **Deployment**: AWS, Render, Vercel
- **Tools**: VS Code, IntelliJ IDEA, Docker, Xilinx Vivado

Soft Skills

- Effective communication and teamwork skills gained through group projects and collaborations.
- Adaptability and quick learning ability in adopting new technologies and methodologies.
- Time management and organizational skills developed through balancing academic and project commitments.
- Open to feedback and use it as a way to grow and improve.

Honors and Awards

- **Dean's List** — University of Moratuwa
 - Awarded for academic excellence in Semester 1, Semester 2, and Semester 3

Voluntary Experience

- **Company Coordinator**, Careers' Day 2025
Department of Computer Science and Engineering, University of Moratuwa
- **Company Coordinator**, CSE Hit the Ground Cricket Competition
Department of Computer Science and Engineering, University of Moratuwa
- **Volunteer**, Sasnaka Sansada, Matale

References

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